DEPI GRAD PROJECT / DATA SCIENCE

E-commerce Customer Intelligence & Retention

Project Documentation | Draft v1.1

PROJECT INFORMATION

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1. Project Planning & Management

1.1 Project Overview

E-commerce Customer Intelligence & Retention is a data-driven analytics and machine learning project designed to transform historical e-commerce transaction data into actionable business insights.

The platform will combine relational database management, SQL-based business analytics, Python-based data processing, supervised machine learning, and interactive data visualization. Its purpose is to help users understand customer purchasing patterns, evaluate sales performance, and explore the potential for predicting future customer behavior.

The project will use the Brazilian E-Commerce Public Dataset by Olist, which contains approximately 100,000 orders from 2016 to 2018 and related information about customers, products, payments, delivery, and reviews [1].

The final machine learning objective will be selected after the team examines the data structure, label feasibility, missing values, class distribution, and prediction time windows.

1.2 Business Problem

E-commerce platforms generate large volumes of transactional data, but raw records alone do not provide a complete understanding of customer value or future purchasing behavior.

Business users need clear answers to questions such as:

- How does revenue vary over time?
- Which products and categories contribute most to sales?
- How frequently do customers place orders?
- What differences exist between one-time and repeat customers?
- Which customer behaviors may help predict future purchases?
- How can historical data support better marketing and retention decisions?

Without a structured analytical process, these questions can be difficult to answer consistently. Incorrect table joins, inconsistent definitions of revenue, missing data, and poorly designed prediction targets can also produce misleading results.

This project will address these challenges by building a connected workflow that moves from data validation and SQL analytics to machine learning outputs and an interactive dashboard.

Important business definition: Unless product costs and other relevant expenses are available, monetary metrics will be described as sales or revenue measures, not profit.

1.3 Project Objectives

The project aims to achieve the following objectives:

1. **Build a structured data foundation:** Inspect the available CSV files, identify primary and foreign keys, establish table relationships, and load relevant data into a relational database.
2. **Improve data quality:** Detect and handle missing values, duplicates, inconsistent data types, invalid records, and other relevant quality issues using documented rules.
3. **Generate business insights:** Use SQL queries and Python analysis to investigate sales trends, order patterns, product performance, customer purchasing frequency, and other relevant business metrics.
4. **Create analytical features:** Develop customer-level and order-level features that accurately represent historical behavior at a clearly defined point in time.
5. **Develop a supervised machine learning model:** Evaluate a suitable classification or regression task. Multi-class prediction and class imbalance will be investigated to align the project with the course requirements.
6. **Evaluate and document model performance:** Compare the selected approach with a baseline, use suitable evaluation metrics, and record relevant experiments.

7. **Deliver an interactive dashboard:** Present understandable charts, business indicators, customer insights, and model outputs through a user-friendly interface.
8. **Prepare an API-based prediction interface:** Subject to time and feasibility, expose the selected model through a FastAPI service and use MLflow to track experiments and model artifacts.

1.4 Scope and Boundaries

In Scope

- Inspection and documentation of the dataset.
- Relational database design and SQL querying.
- Data validation and quality assessment.
- Exploratory data analysis (EDA).
- Customer and sales analytics.
- Feature engineering and preprocessing.
- Supervised machine learning.
- Multi-class classification and class-imbalance investigation, where supported by the data.
- Hyperparameter optimization using Bayesian Search, where feasible.
- Model evaluation and experiment tracking with MLflow.
- Dashboard development using Streamlit or another agreed Python-based tool.
- Basic model serving through FastAPI, subject to successful implementation.
- Documentation, testing, version control, and final presentation.

Out of Scope

- Integration with a live commercial e-commerce system.
- Real-time transactional processing.
- Deployment to Microsoft Azure or another mandatory cloud platform.
- Docker-based deployment.
- Production-grade marketing automation.
- Claims about causal impact on customer behavior without an appropriate experimental design.
- Profit prediction when reliable cost and expense data are unavailable.

Clustering and advanced customer segmentation may be considered as an extension, but they are not a substitute for the required supervised machine learning component.

1.5 Proposed Methodology and Architecture

The project will follow a staged analytical workflow:

Stage 1 — Data Understanding and Collection

Inspect the supplied dataset files, record table sizes and column definitions, identify data types and keys, and document the data source and limitations.

Stage 2 — Database Design and SQL Analytics

Create the required relational tables, define relationships, validate joins, and build SQL queries for business metrics. Tables will be joined according to their actual level of detail. For example, order-item records and payment records may contain multiple rows per order; they should be aggregated appropriately before being combined when necessary.

Stage 3 — Data Preparation and EDA

Use Python and Pandas to investigate data quality, missingness, unusual values, distributions, relationships, and patterns. Data-cleaning decisions must be documented rather than applied automatically without justification.

Stage 4 — Feature Engineering and Target Definition

Define an observation period, a prediction cutoff, and a future outcome window. Create features using only information available before the prediction cutoff. Exclude records that do not have sufficient future observation time when constructing future-purchase targets.

Stage 5 - Machine Learning

Establish a simple baseline, train candidate models, tune selected hyperparameters, handle imbalance where appropriate, and assess performance using a suitable validation strategy. A time-based split should be considered because the records are historical transactions.

Stage 6 - Experiment Tracking and API

Use MLflow to record model parameters, metrics, and artifacts where practical [5]. If the API extension is implemented, FastAPI can validate incoming prediction requests and expose model results through documented endpoints [6].

Stage 7 - Dashboard and Final Validation

Use Streamlit or an agreed alternative to display business metrics, trends, and model outputs [7]. Test the dashboard against validated SQL or Python results and ensure that its labels and definitions are understandable.

Proposed Architecture

Raw CSV Files → Data Inspection and Cleaning → Relational Database → SQL Analytics → Python Feature Engineering → Model Training and Evaluation → MLflow Tracking → FastAPI Prediction Interface (optional) → Interactive Dashboard

The dashboard may connect directly to prepared analytical tables, a local prediction pipeline, or the API. The final integration design will depend on the team's implementation and time constraints.

1.6 Team Roles and Responsibilities

The following allocation defines the team's current division of responsibilities. Members should coordinate on shared dependencies and review each other's work.

Team Member	Role	Main Responsibilities
Karas Ebrahim Ghaly	Project Lead, SQL, Relationships, Deployment	SQL schema, preparing the ML dataset, and GitHub coordination.
Mariam Nabil Ali	ML Dataset Pre-review & Business Analytics	Dataset evaluation, schema auditing, and supporting business logic definition.
Toqa Tawfiq Abdel-Moez	Data Cleaning, EDA & Data Preprocessing	Handling outliers, checking duplicates, correlation analysis, and data visualization.
Mariam Mohammed Elsayed	Machine Learning & Model Evaluation	Target definition, building baselines and candidate models, model evaluation, and interpreting limitations.
Mina Raoof Abdel-Malak	Testing, Project Integration & Dashboards	Streamlit dashboard development, system integration, end-to-end testing, and presentation prep.

Team Collaboration Principles

- Each major task should have one accountable owner and at least one reviewer.
- The team should agree on shared definitions before parallel development begins.
- SQL results, feature definitions, and model evaluation outputs should be reproducible.
- Code, documentation, and decisions should be version-controlled where practical.
- Integration should begin before the final week rather than being postponed until every component is complete.

1.7 Project Timeline and Gantt-Style Plan

The project delivery window is seven weeks, from 19 October to 6 December 2026. The weekly plan below maps the main activities to that fixed window and should be aligned with any official DEPI checkpoints.

Activity	W1	W2	W3	W4	W5	W6	W7
Project planning and business questions	■						
Dataset inspection and schema design	■	■					
Database setup and SQL analytics		■	■				
Data cleaning and EDA		■	■	■			
Feature engineering and target definition			■	■			
Baseline model and candidate models				■	■		
Tuning, evaluation, and model tracking					■	■	
Dashboard development				■	■	■	
API and component integration						■	■
Testing and documentation						■	■
Final review and presentation							■

Legend: ■ = planned activity.

Week ranges: W1 19-25 Oct; W2 26 Oct-1 Nov; W3 2-8 Nov; W4 9-15 Nov; W5 16-22 Nov; W6 23-29 Nov; W7 30 Nov-6 Dec.

Some activities intentionally overlap. For example, preliminary dashboard development can begin while the model is being developed, using validated sample results until the final model output becomes available.

1.8 Milestones and Deliverables

Milestone	Expected Deliverable	Completion Evidence
M1 - Planning Approval	Project plan, problem statement, and initial requirements	Team review and documented decisions
M2 - Data Foundation	Data dictionary, relational schema, and loaded tables	Row counts, key checks, and successful SQL queries
M3 - Analytics Ready	Cleaning report, EDA, and initial business metrics	Reproducible analysis and documented assumptions
M4 - Modeling Ready	Feature dataset and agreed prediction target	Leakage checks, target definition, and label-distribution report
M5 - Model Evaluated	Baseline, candidate model, and evaluation report	Metrics on an appropriate validation/test split
M6 - Application Integrated	Dashboard and, if completed, API	Demonstrable end-to-end workflow
M7 - Final Submission	Documentation, code, results, and presentation	Repository review and final team sign-off

1.9 Risk Management

Risk	Likelihood	Impact	Mitigation Strategy
Missing or inconsistent data	High	High	Profile all tables, document quality issues, and establish repeatable cleaning rules.
Incorrect joins or duplicated totals	Medium	High	Check table grain, validate key uniqueness, and reconcile aggregates before and after joins.
Target leakage in the ML dataset	Medium	High	Define a prediction cutoff and ensure features use only information available before it.

Risk	Likelihood	Impact	Mitigation Strategy
Severe class imbalance	High	High	Inspect label distribution, use suitable metrics, compare class-weighting or resampling approaches, and retain a simple baseline.
Weak predictive performance	Medium	High	Compare several reasonable approaches, review feature quality, and report limitations honestly.
Team dependencies or schedule delays	Medium	Medium	Assign owners and reviewers, track milestones, and integrate components incrementally.
Dashboard and model integration issues	Medium	Medium	Define output formats early and test the integration with a small example.
Scope expansion	Medium	Medium	Prioritize required deliverables and defer optional features until the core workflow works.
Inconsistent business definitions	Medium	High	Maintain a shared metric dictionary and agree on revenue, order status, and customer definitions.

1.10 Key Performance Indicators

Success will be assessed through data quality, analytical correctness, model evaluation, application usability, and completion of the project deliverables.

Area	Proposed KPI	Measurement Method
Data Preparation	Quality issues are identified and documented	Data-quality report covering missing values, duplicates, types, and invalid records
Database	Required tables and relationships are usable	Key checks, join validation, and successful execution of planned queries
Business Analytics	Core business metrics are reproducible	Reconcile dashboard metrics with approved SQL or Python outputs
Feature Engineering	Features respect the prediction cutoff	Documented leakage review and time-window validation
Machine Learning	Candidate model improves on a defined baseline	Comparison using task-appropriate metrics on held-out data
Imbalanced Classification	Minority-class performance is assessed	Macro F1, per-class precision/recall, and confusion matrix; PR-AUC where appropriate
Experiment Tracking	Relevant training runs are recorded	Review saved parameters, metrics, and artifacts in MLflow
Dashboard	Required indicators and model outputs are accessible	Functional testing against a defined dashboard checklist
API, if implemented	Valid requests produce structured responses	Endpoint tests, schema validation, and error-case checks
Project Management	Milestones and responsibilities are tracked	Milestone review and documented completion status

Numerical performance targets should be confirmed after the baseline, data characteristics, and model objective are known. A high overall accuracy alone will not be considered sufficient evidence of model quality when class imbalance is present.

2. Literature Review

2.1 Review Objective

This review examines related research, analytical methods, and existing technical solutions relevant to customer purchase prediction, customer-value analysis, transaction analytics, and the delivery of machine learning insights through an application.

The review is intended to inform the project's design rather than to copy an existing implementation. Academic research, dataset documentation, public project repositories, and official technical documentation are considered different types of evidence and should not be treated as interchangeable.

2.2 Related Research and Existing Solutions

2.2.1 Brazilian E-Commerce Public Dataset by Olist

The Olist dataset provides anonymized historical information from approximately 100,000 Brazilian e-commerce orders between 2016 and 2018. Its related tables cover several dimensions of an order, including customers, items, payments, delivery, products, and reviews [1].

Relevance to this project: The relational structure makes it suitable for demonstrating database design, SQL aggregation, data cleaning, customer-level feature engineering, business analysis, and supervised learning.

Advantages

- Represents historical commercial transactions rather than a small artificial example.
- Contains numerical, categorical, and date-related information.
- Supports several analytical perspectives through related tables.
- Enables realistic practice with joins, aggregation, and data-quality decisions.

Limitations

- The data describes a historical period and does not represent a live e-commerce environment.
- The team must verify the completeness and suitability of the available observation windows for its chosen target.
- Relationships among orders, payments, and order items require careful treatment to avoid duplicate counting.
- Model findings may not generalize to a different business, country, or time period.

The dataset is the project's data foundation, not a predictive model or a complete customer intelligence solution by itself.

2.2.2 Customer Purchase Prediction in B2C E-Business

Chen, Xu, Xu, and Gou published a systematic review of customer purchase prediction in *Expert Systems with Applications* in October 2024. The study examined 98 articles published between 1995 and 2023 and discussed the field's research directions and future agenda [2].

Relevance to this project: The review supports treating purchase prediction as a distinct research problem that requires explicit target definitions, appropriate data, and evaluation rather than assuming that a model trained on historical transactions will automatically predict customer behavior effectively.

Advantages

- Provides a broad overview of a research area rather than focusing on one dataset.
- Offers academic context for selecting and motivating a purchase-prediction problem.
- Helps frame the project's assumptions and evaluation decisions.

Limitations

- A systematic review does not directly supply a ready-to-deploy model for the Olist dataset.
- The findings of individual studies may depend on their datasets, business contexts, feature definitions, and prediction windows.
- The team must still validate its own target and candidate methods empirically.

2.2.3 RFM and Extended Customer Behavior Analysis

Recency, Frequency, and Monetary (RFM) analysis describes customers using how recently they purchased, how frequently they purchase, and how much they have spent. An article in the *Journal of Retailing and Consumer Services* introduced an extended RFMT framework by adding interpurchase time to examine shopping regularity and customer behavior over time [3].

Relevance to this project: RFM-like features may provide interpretable baselines for customer analytics and can help the team summarize historical purchasing patterns before attempting more complex supervised prediction.

Advantages

- Uses business-friendly and interpretable features.
- Can support customer profiling and descriptive segmentation.
- Provides a useful reference point for constructing customer-level features.

Limitations

- Historical behavior alone does not guarantee accurate predictions of future outcomes.
- Results depend on the observation period, customer eligibility rules, and transaction definitions.
- Segmentation is not equivalent to a supervised prediction model; clustering alone would not satisfy the project's supervised-learning objective.

RFM-derived measures may be considered as input features or descriptive analyses. Their usefulness must be tested against the chosen prediction target.

2.2.4 Existing Public Olist Retention-Prediction Project

A public GitHub repository describes an end-to-end Olist retention and repeat-purchase analysis using SQL, Python, customer analytics, feature engineering, and supervised learning [4].

Relevance to this project: It provides a practical example of how an Olist-based analytical workflow can connect transaction preparation with a prediction problem.

Advantages

- Uses the same general dataset family.
- Illustrates a workflow that includes data preparation, analytical features, and model evaluation.
- Can help the team identify implementation questions and potential pitfalls.

Limitations

- It is an individual public project, not a peer-reviewed benchmark or an independently verified business deployment.
- Its target definition, data-preparation choices, splits, and results may differ from this project.
- Its reported results should not be assumed to be reproducible or transferable without independent verification.

The team should use public implementations as learning references, document its own methodology, and generate its own results rather than copying reported performance claims.

2.2.5 Tools for Model Tracking and Application Delivery

MLflow provides facilities for recording experiment parameters, metrics, code-related information, and artifacts, helping teams compare model runs and preserve experiment history [5]. FastAPI supports structured request bodies, data validation, and automatically generated API documentation [6]. Streamlit provides a Python-based framework for developing interactive data applications [7].

These tools do not establish that a model is accurate. They support the engineering workflow surrounding experimentation, serving, and communication.

Advantages

- Improve reproducibility and comparison of training runs.
- Make model outputs accessible through a structured API when needed.
- Provide an accessible interface for presenting business metrics and predictions.

Limitations

- They require integration, testing, and consistent data contracts.
- An API cannot compensate for an invalid target or leakage-prone training dataset.

- A dashboard can communicate misleading findings if metric definitions and data checks are incorrect.

2.3 Comparative Analysis

Approach	Main Purpose	Strengths	Main Limitations	Role in This Project
SQL Business Analytics	Summarize sales, orders, and product performance	Transparent, auditable, and well suited to relational data	Primarily descriptive unless combined with prediction methods	Core requirement
RFM / Extended RFM	Describe customer purchase patterns	Interpretable and business-friendly	Does not independently predict future outcomes	Candidate feature framework and descriptive analysis
Supervised Purchase Prediction	Estimate a future customer outcome	Produces measurable predictions that can be evaluated	Sensitive to target design, label availability, leakage, and imbalance	Main ML component
Existing Olist Public Project	Illustrate an end-to-end analytical workflow	Practical implementation reference	Results and design choices may not transfer	Supporting reference only
MLflow + FastAPI + Streamlit	Track experiments, serve predictions, and present results	Helps connect experimentation with usable outputs	Requires additional integration and testing	Advanced implementation, subject to feasibility

2.4 Identified Gaps and Project Contribution

Based on the sources reviewed, the project will focus on bringing several components together into one coherent workflow rather than claiming to invent a new prediction algorithm.

The intended contribution is to:

1. Connect relational data analysis with customer-level behavioral features.
2. Define business metrics consistently and validate them against source tables.
3. Establish a defensible prediction cutoff and future-outcome window.
4. Investigate class imbalance and evaluate more than overall accuracy when classification is selected.
5. Compare a baseline with candidate models using appropriate validation.
6. Present findings and limitations in a usable dashboard.
7. Document the workflow so that other team members can understand and reproduce the principal results.

The final project contribution will depend on the target selected, the data-quality findings, and the empirical results. No predictive performance or business improvement is claimed at the planning stage.

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Main Interest	Primary Needs
Course Instructor / Evaluator	Assess technical implementation and project outcomes	Clear objectives, evidence of required methods, correct evaluation, and complete documentation
Project Team	Develop and integrate the system	Defined responsibilities, shared metric definitions, accessible code, and manageable milestones
Business Analyst	Understand sales and customer behavior	Reliable metrics, clear trends, and interpretable summaries
Marketing or Customer-Relations User	Identify useful customer behavior patterns	Customer insights and appropriately qualified predictions to support campaign planning
Dashboard User	Explore results without writing code	Understandable charts, filters, clear labels, and explanatory descriptions
Technical Maintainer	Run, test, and update the application	Reproducible setup, documented interfaces, validation, and readable code

The dataset is historical and anonymized. Stakeholder needs are therefore represented as project use cases; the system is not assumed to be connected to real customers or a live commercial operation.

3.2 User Stories

The following user stories describe the intended functionality and can be refined after the team reviews the dashboard design.

US-01 — Sales Overview

As a business analyst, I want to view sales and order trends over time so that I can understand changes in recorded business activity.

US-02 — Product Performance

As a business analyst, I want to compare product categories using clearly defined sales metrics so that I can identify categories with stronger recorded performance.

US-03 — Customer Behavior

As a customer analyst, I want to examine purchasing frequency, recency, and spending patterns so that I can understand differences in historical customer behavior.

US-04 — Data Quality

As a data scientist, I want to review missing values, invalid records, and data-type issues so that I can justify preparation decisions.

US-05 — Customer Prediction

As a business user, I want to review the model's estimated customer outcome so that I can explore how historical information may support future planning.

US-06 — Model Evaluation

As a project evaluator, I want to see performance metrics and a comparison with a baseline so that I can judge whether the model adds measurable predictive value.

US-07 — Filtered Analysis

As a dashboard user, I want to filter available analysis by supported dimensions such as time period, category, or order status so that I can investigate specific subsets of the data.

US-08 — Reproducible Experiments

As a project team member, I want model runs and evaluation results to be recorded so that I can compare experiments and understand which configuration produced a result.

US-09 — Prediction Interface

As a technical user, I want a structured prediction endpoint, if the API extension is implemented, so that the model can be tested independently of the dashboard.

3.3 Main Use Cases

UC-01: Explore Business Metrics

Primary actor: Business analyst **Precondition:** Validated analytical data is available.

Main flow:

1. The user opens the dashboard.
2. The system displays available summary metrics.
3. The user selects a supported date range or analytical filter.
4. The dashboard updates the relevant charts and tables.
5. The user reviews the resulting trends.

Expected outcome: The user can explore the selected metrics and their definitions without directly querying the database.

UC-02: Investigate Customer Behavior

Primary actor: Customer analyst

Main flow:

1. The user selects the customer analysis section.
2. The system displays approved customer-level measures.
3. The user examines recency, frequency, and monetary patterns where available.
4. The system presents the analytical results with clear labels and relevant limitations.

Expected outcome: The user gains a descriptive view of historical customer behavior.

UC-03: Review Model Performance

Primary actor: Data scientist or evaluator

Precondition: At least one model and baseline have been evaluated.

Main flow:

1. The user opens the model evaluation section.
2. The system displays the selected target and validation approach.
3. The user reviews appropriate metrics, class distribution, and model comparisons.
4. The user reviews documented limitations.

Expected outcome: Model quality is presented using evidence rather than an unsupported claim of success.

UC-04: Obtain a Prediction

Primary actor: Dashboard user or API client **Precondition:** A valid model has been selected and the required inputs are available.

Main flow:

1. The user supplies the required input data or selects an eligible customer record.
2. The system validates that the inputs meet the model’s requirements.

3. The model generates the prediction.
4. The system returns a structured result with an appropriate explanation.

Alternative flow: If the input is invalid, unavailable, or outside the supported schema, the system returns a clear error message instead of silently generating a result.

Expected outcome: A prediction is returned only when the necessary inputs and model artifacts are available.

3.4 Functional Requirements

ID	Requirement	Priority
FR-01	The system shall load and inspect the relevant source data files.	Must
FR-02	The project shall document table columns, key fields, data types, and relevant relationships.	Must
FR-03	The database layer shall support the agreed business-analysis queries.	Must
FR-04	The team shall validate important joins and aggregation rules to reduce duplicate counting.	Must
FR-05	The data-preparation workflow shall identify and document relevant missing values, duplicates, invalid records, and type inconsistencies.	Must
FR-06	The project shall provide exploratory analysis of relevant sales, product, and customer behavior.	Must
FR-07	The team shall construct an ML-ready dataset with a documented target and feature definitions.	Must
FR-08	The team shall train and evaluate at least one supervised model appropriate to the finalized target.	Must
FR-09	The team shall compare candidate-model performance with a defined baseline.	Must
FR-10	When classification is selected, the evaluation shall consider class distribution and use appropriate metrics beyond overall accuracy where necessary.	Must
FR-11	The application shall display the agreed business metrics and relevant analytical findings.	Must
FR-12	Model results displayed in the application shall correspond to documented model outputs and evaluation results.	Must
FR-13	The project should record model parameters, metrics, and relevant artifacts in MLflow.	Should
FR-14	The project should provide a FastAPI prediction endpoint if time and integration feasibility permit.	Should
FR-15	The application should present clear explanations of metrics and model limitations.	Should

3.5 Non-Functional Requirements

Performance

- Common dashboard interactions should complete within a reasonable time on the team's development hardware.
- The team should measure actual response times rather than claim performance without testing.
- Repeated expensive calculations should be reviewed for opportunities to cache or precompute results.
- Any API performance target should be finalized after the initial implementation and measured under documented test conditions.

Security and Privacy

- The application should not expose database credentials, secret keys, or environment-specific configuration in the public repository.
- Database access should use appropriate permissions for the local development setup.
- Prediction requests should be validated before they are processed.
- Data and outputs should be handled according to the anonymized dataset's intended use and the course's requirements.
- Error messages should not unnecessarily disclose internal configuration or sensitive operational details.

Usability

- Dashboard pages should use understandable titles, labels, units, and chart legends.
- Business terms such as orders, customers, sales, and average order value should have documented definitions.
- Filters should behave consistently and make the selected criteria apparent.
- Model outputs should be presented with a short explanation of what the prediction represents and what it does not establish.
- Invalid inputs or unavailable results should produce clear feedback.

Reliability and Maintainability

- The project should provide a documented method for preparing the data and running the main components.
- Important data transformations and modeling decisions should be reproducible.
- The team should test critical SQL queries, target construction, model evaluation, and application behavior.
- The project should use version control and maintain a clear repository structure.
- Model or data changes should trigger appropriate validation before results are treated as final.
- The documentation should distinguish implemented features from planned or optional features.

Reproducibility

- Random seeds should be recorded where relevant.
- The train-validation-test strategy should be documented.
- Feature definitions and target-construction rules should be version-controlled.
- When MLflow is implemented, relevant parameters, metrics, and artifacts should be recorded for meaningful runs.

3.6 Data and Machine Learning Requirements

Data Requirements

The source data should be inspected before the team commits to the final analytical model. At minimum, the team should document:

- Table row counts and column definitions.
- Key uniqueness and relationship validity.
- Missing-value patterns and relevant invalid records.
- The meaning and status of orders used in each metric.
- Potential duplicate counting from one-to-many relationships.
- The observation period and the completeness of future outcome windows.
- The definitions of customer, order, item, payment, and monetary measures.

Not every available source table must be included in every query or model. The selection should follow the business question and the grain of the analytical dataset.

Machine Learning Target

The final target is **not yet confirmed**. Candidate objectives include:

1. **Repeat-purchase prediction:** Estimate whether a customer will place a subsequent order within a defined future period.
2. **Future-order activity classification:** Classify a customer's future order activity into meaningful classes, such as zero, one, or multiple subsequent orders, if sufficient observations support those classes.
3. **Customer-value regression:** Estimate a defined future monetary measure, provided it can be constructed reliably from the data.

A multi-class target may be considered to align with the course's advanced requirements, but it must represent a meaningful business outcome and have enough examples for evaluation. If some classes are extremely rare, the team must document the issue and reconsider the class definitions or modeling approach.

Leakage Prevention

- Features must not contain information that would only become available after the prediction cutoff.
- The target outcome must be calculated from a future window separate from the feature observation period.
- Records lacking sufficient follow-up time should be excluded where necessary.
- A time-based split should be considered to better reflect the task of predicting future outcomes from historical data.
- Any preprocessing fitted to training data must not inadvertently use information from the held-out test data.

Model Evaluation

The evaluation plan must match the selected task.

For classification, the team should consider the confusion matrix, per-class precision and recall, macro F1, and suitable probability-ranking metrics such as PR-AUC where appropriate. Balanced accuracy may also be useful. Overall accuracy should not be the only metric when classes are imbalanced.

For regression, the team should use suitable error measures such as MAE and RMSE and compare the results with a simple baseline. The meaning and units of the target must be stated clearly.

Bayesian hyperparameter optimization can be considered after a valid baseline and a reliable evaluation setup are established. Model complexity should be justified by evidence rather than used only to make the project appear more advanced.

3.7 Acceptance Criteria

The project will be considered ready for final review when the following agreed criteria are met:

- ☐ The source dataset and its limitations are documented.
- ☐ Required tables have been inspected and the relevant database relationships are documented.
- ☐ Core SQL queries execute correctly and important aggregations have been validated.
- ☐ Data-cleaning decisions and exploratory findings are documented.
- ☐ The ML target, observation period, outcome window, and eligible records are clearly defined.
- ☐ Leakage risks have been reviewed and the evaluation split is documented.
- ☐ At least one supervised model is evaluated against a baseline.
- ☐ Evaluation metrics are appropriate to the task and address class imbalance where relevant.
- ☐ The dashboard presents validated metrics and accurately labeled model outputs.
- ☐ Optional components, including MLflow and FastAPI, are clearly identified as implemented or not implemented.
- ☐ The repository contains the agreed documentation and instructions needed to review the work.
- ☐ The team can demonstrate the main workflow during the final presentation.

3.8 Open Decisions for Team Confirmation

The following items remain open until the team has discussed them and reviewed the data:

1. **Final ML objective:** Select classification or regression based on business value, data feasibility, and course requirements.
2. **Multi-class design:** Decide whether a meaningful multi-class target can be supported with sufficient examples in each class.
3. **Prediction horizon:** Agree on the observation period and future outcome window.
4. **Database technology:** Select the database engine based on team familiarity and course constraints.
5. **Application integration:** Decide whether the dashboard will run predictions locally or call a FastAPI endpoint.
6. **Project schedule:** Confirm weekly checkpoints within the agreed 19 October-6 December 2026 window.
7. **Advanced features:** Confirm the time available for Bayesian Search, MLflow, and API integration after the core workflow is functional.

Any lecturer feedback or formal grading criteria should be added after the actual feedback or official rubric is available. These should not be presented as completed assessments before they have been provided.

4. References

[1] Olist. *Brazilian E-Commerce Public Dataset by Olist*. Kaggle. Dataset description covering approximately 100,000 orders from 2016–2018. https://www.kaggle.com/olistbr/brazilian-ecommerce

[2] Chen, S., Xu, Z., Xu, D., & Gou, X. (2024). *Customer purchase prediction in B2C e-business: A systematic review and future research agenda*. Expert Systems with Applications, 252, Article 124261. https://doi.org/10.1016/j.eswa.2024.124261

[3] *Customer segmentation by web content mining*. (2021). Journal of Retailing and Consumer Services, 61, Article 102588. Research describing an extended RFM framework incorporating interpurchase time. https://doi.org/10.1016/j.jretconser.2021.102588

[4] bori-dori. *olist-retention-prediction*. Public GitHub repository describing an Olist-based retention and repeat-purchase prediction workflow. Used as an implementation reference, not as a peer-reviewed benchmark. https://github.com/bori-dori/olist-retention-prediction

[5] MLflow. *MLflow Tracking Documentation*. Official documentation for recording model parameters, metrics, and artifacts. https://mlflow.org/docs/latest/ml/tracking/

[6] FastAPI. *Request Body and Validation Documentation*. Official documentation for structured request bodies, validation, and generated API schemas. https://fastapi.tiangolo.com/tutorial/body/

[7] Streamlit. *Streamlit Documentation*. Official documentation for building interactive Python-based data applications. https://docs.streamlit.io/

Document Status: Draft v1.1 - For Team Discussion and Revision. Next Step: Confirm the final ML objective, prediction window, database choice, and implementation priorities with the team before approving the document.